

# Black body radiation

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

| QUESTION |                                                                                             | ANSWER                                                                                   | ✓                        |
|----------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------|
| 1        | Name the type of radiation that all objects emit.                                           | Infrared radiation                                                                       | <input type="checkbox"/> |
| 2        | Describe how the infrared radiation emitted by an object depends on its temperature.        | The hotter the object, the more it emits.                                                | <input type="checkbox"/> |
| 3        | Define a perfect black body.                                                                | An object that absorbs all the radiation that falls on it.                               | <input type="checkbox"/> |
| 4        | State how an object's ability to absorb radiation relates to its ability to emit radiation. | A good absorber is also a good emitter.                                                  | <input type="checkbox"/> |
| 5        | Describe how the radiation emitted by an object changes as its temperature increases.       | The intensity increases and the peak moves to a shorter wavelength.                      | <input type="checkbox"/> |
| 6        | An object is at a constant temperature. Compare the radiation it emits and absorbs.         | It absorbs and emits radiation at the same rate.                                         | <input type="checkbox"/> |
| 7        | State when the temperature of an object increases, in terms of radiation.                   | When it absorbs radiation faster than it emits it.                                       | <input type="checkbox"/> |
| 8        | State what all objects do, whatever their temperature.                                      | They emit and absorb infrared radiation.                                                 | <input type="checkbox"/> |
| 9        | State when the temperature of an object decreases, in terms of radiation.                   | When it emits radiation faster than it absorbs it.                                       | <input type="checkbox"/> |
| 10       | Give three factors that the temperature of the Earth depends on.                            | The rates at which it absorbs and emits radiation, and how much is reflected into space. | <input type="checkbox"/> |